Task 5 Web server deployement using APACHE

Procedure:

Step 1: Verify Apache Installation

Checked whether Apache Web Server was installed on the system. (In built installed in MacOS)

Command: httpd -v

Step 2: Obtain Administrative Privileges

Switched to the root user to perform server administration tasks.

Command: sudo -I

Step 3: Verify Apache Configuration

Checked Apache configuration files for syntax errors.

Command: apachectl configtest

Step 4: Start Apache Web Server

Command: apachectl -k start

Step 5: Open Apache Document Root Directory and create webpage

Command: /Library/WebServer/Documents/index.html

Command: nano /Library/WebServer/Documents/index.html

Step 6: Restart Apache

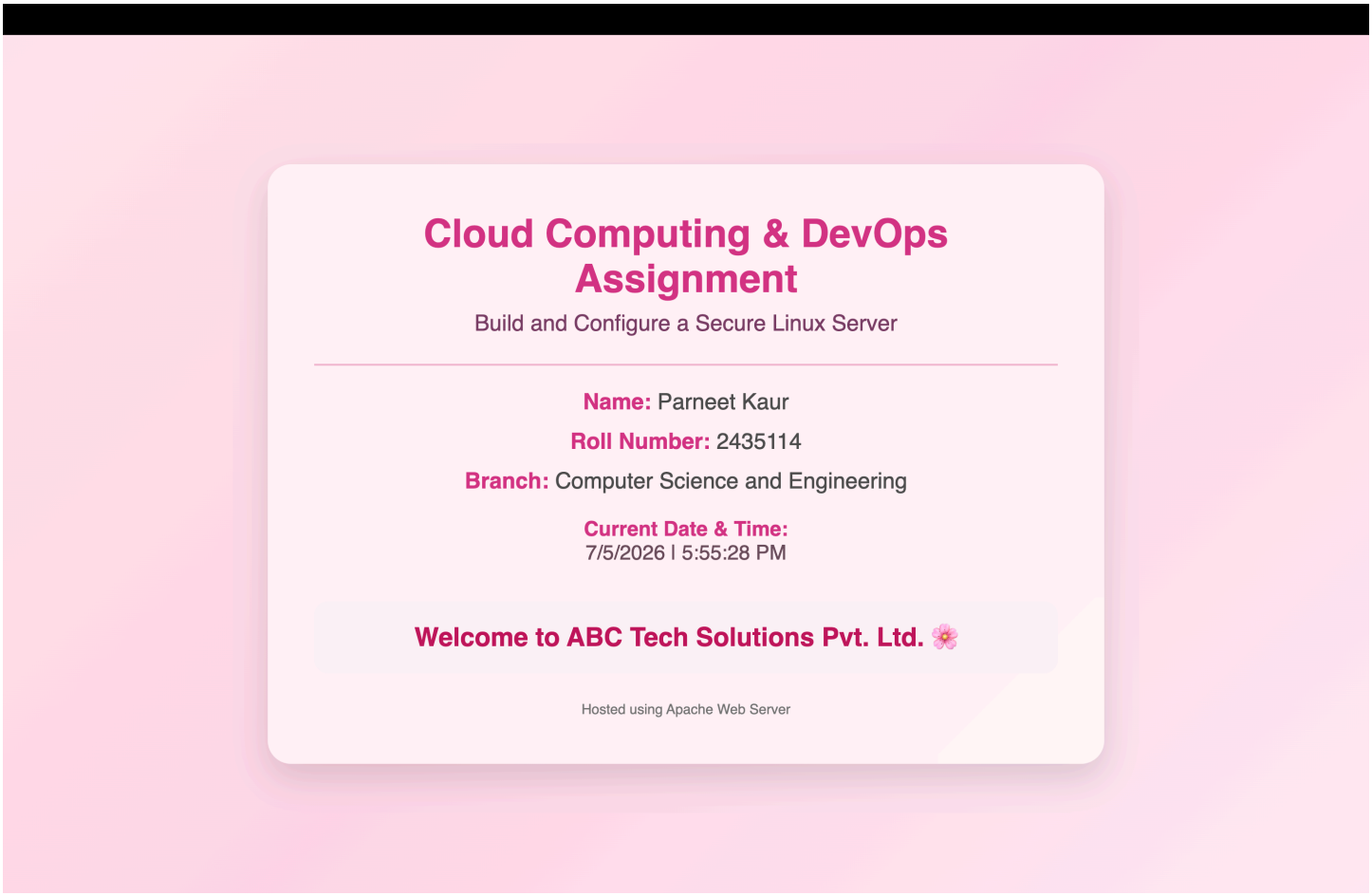
Command: apachectl restart

Step 7: Access the Webpage Using Localhost

Command: http://localhost

Step 8: Access the Webpage Using Server IP

Command: http://IP



Task 6 Bash automation script

```
[Parneets-MacBook-Air:~ root# nano server_health.sh
[Parneets-MacBook-Air:~ root# chmod +x server_health.sh
[Parneets-MacBook-Air:~ root# ./server_health.sh
        SERVER HEALTH REPORT

 Current Date & Time:
 Mon Jul  6 10:10:28 IST 2026

 Logged-in User:
 root

 Hostname:
 Parneets-MacBook-Air.local

 IP Address:
 172.16.11.8

 Disk Usage:
 Filesystem        Size    Used   Avail Capacity iused ifree %iused  Mounted on
 /dev/disk2s1s1   228Gi    12Gi   129Gi     9%    459k  1.4G    0%   /

 Memory Usage:
 Mach Virtual Memory Statistics: (page size of 16384 bytes)
 Pages free:                                11973.
 Pages active:                             142119.
 Pages inactive:                           140937.
 Pages speculative:                           516.
 Pages throttled:                               0.
 Pages wired down:                          86656.
 Pages purgeable:                           17536.
 "Translation faults":                   27040449.
 Pages copy-on-write:                      570521.
 Pages zero filled:                      39956759.
 Pages reactivated:                       3643125.
 Pages purged:                             612258.
 File-backed pages:                         97438.
 Anonymous pages:                          186134.
 Pages stored in compressor:               295616.
 Pages occupied by compressor:             104570.
 Decompressions:                          1876403.
 Compressions:                            3026939.
 Pageins:                                 1949211.
 Pageouts:                                  25209.
 Swapins:                                       0.
 Swapouts:                                      0.

 CPU Load:
 10:10  up 21:02, 3 users, load averages: 1.79 1.88 2.26

 System Uptime:
 10:10  up 21:02, 3 users, load averages: 1.79 1.88 2.26

        REPORT COMPLETED
 Parneets-MacBook-Air:~ root#
```

```
#!/bin/bash

echo "        SERVER HEALTH REPORT"

echo ""
echo "Current Date & Time:"
date

echo ""
echo "Logged-in User:"
whoami

echo ""
echo "Hostname:"
hostname

echo ""
echo "IP Address:"
ipconfig getifaddr en0

echo ""
echo "Disk Usage:"
df -h /

echo ""
echo "Memory Usage:"
vm_stat

echo ""
echo "CPU Load:"
uptime

echo ""
echo "System Uptime:"
uptime

echo ""
echo "        REPORT COMPLETED"
```